

**APPROVED METHOD:**  
**LIFE TESTING OF HIGH INTENSITY**  
**DISCHARGE (HID) LAMPS**  
AN AMERICAN NATIONAL STANDARD

**ANSI/IES LM-47-20(R2023)**

**APPROVED METHOD:  
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HIGH INTENSITY DISCHARGE (HID) LAMPS  
AN AMERICAN NATIONAL STANDARD**

Publication of this document  
has been approved by the IES.  
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**Prepared by:  
The IES Testing Procedures Committee**



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## Foreword

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This Approved Method is a revision of *IES LM-47-2012, IES Approved Method: Life Testing of High Intensity Discharge (HID) Lamps*; changes have been made to update information.

## 1.0 Introduction and Scope

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### 1.1 Introduction

This document describes the procedures by which high intensity discharge (HID) lamps can be operated under controlled conditions to obtain optimally comparable data on individual lamp life, changes in light output, and other parameters that vary during the life of the lamp.

HID lamps are characterized as light producing discharge devices whose discharge containment envelopes have a surface loading more than three watts per square centimeter. Additional characteristics of these lamp types are available.<sup>1-4</sup>

Light is produced in a high intensity discharge lamp by passing current through a vapor at relatively high pressure. Arc tube pressures for HID lamps can range from just less than one to as much as five atmospheres during lamp operation.

In a high pressure sodium lamp (HPS) the arc discharge passes through combined vapors of mercury and sodium. The sodium radiation dominates the color appearance of the light produced, which is characteristically gold and contains a minimal ultraviolet energy output.

Light is produced in the metal halide (MH) lamp by an electrical discharge through combined vapors of mercury and other metals. These metals have the effect of increasing the light output in the blue, green, and yellow/red bands of the visual spectrum. Due to the spectral energy distribution in the lamp output from the metal vapors, the color appearance of MH HID lamps is white. In addition to visible light, ultraviolet energy is also produced in the arc and is used in phosphor coated lamps to improve color characteristics.

In mercury vapor lamps, light is generated by electric discharge through a vapor of mercury. The light produced is bright white, often with the blue green cast. The color is dramatically improved by adding a phosphor coating to the outer bulb.

It is often important to know the light output, efficacy, and lamp lumen maintenance along with the life of these lamps. The annex to this document discusses this issue. (For information on the photometry of HID lamps, refer to *ANSI/IES LM-51-20, Approved Method: Electrical and Photometric Measurements of High Intensity Discharge Lamps*.<sup>5</sup>)

HID lamps are affected by many variables such as operating cycle, conditions imposed by auxiliary equipment, ambient temperature, vibration, airflow, and orientation. Test conditions and programs should be designed to give comparable results when adopted by various laboratories. The recommendations of this IES approved method have been made with these objectives in mind.

### 1.2 Scope

This Approved Method describes the procedures to be followed and the precautions to be observed in obtaining uniform and reproducible measurements during life testing of high intensity discharge lamps under standard conditions. This Approved Method includes other technologies within the metal halide family including ceramic metal halide and pulse start metal halide lamps. It addresses life testing of high intensity discharge lamps operated on auxiliary devices, either external or integrated, designed and certified to meet lamp industry standards and tolerance. Xenon arc lamps, low pressure sodium lamps, and automotive lamps are not addressed in this IES Approved Method.

## 2.0 Normative References

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### 2.1 LS-1-20

Illuminating Engineering Society. Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020. Online: [www.ies.org/standards/definitions/](http://www.ies.org/standards/definitions/). (Accessed 2019 Mar 20).

## 3.0 Nomenclature and Definitions

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The units of electrical measurement used in this test method are the volt, the ampere, and the watt.

The terms used in this document follow the definitions given in LS-1-20 (see **Section 2.1**).

### 3.1 auxiliary equipment

The reactor or ballast and, if required, starting device needed for the proper operation of discharge lamps. The auxiliary equipment is used to limit lamp current and provide proper starting voltage and is essential because discharge lamps have a negative volt-ampere characteristic.

*Note:* For lamps with integrated auxiliary equipment (self ballasted), subsequent references to auxiliary equipment can be ignored.

### 3.2 lamp failure

The point where a lamp stops operating. Lamp failures from manufacturing defects are reported but not included in the calculation of lamp life.

### 3.3 life of a test lamp

The actual operating time of the lamp until failure, expressed in hours, and which shall not include any off time.

### 3.4 life test rack

A suitable framework or supporting media where multiple lamp testing positions are available.

### 3.5 rated lamp life

The life value assigned to a particular type lamp. This is commonly a statistically determined estimate of median operational life.

*Note:* For life rating, the applicable definition of median is “the total operating time at which under normal operating conditions 50% of any large group of initially installed lamps is expected to be still operating.”

### 3.6 regulation

The constancy of the voltage (and/or current) applied to the equipment under test.

### 3.7 seasoning

The process of operating a new lamp for a fixed period of time. This is sometimes referred to as aging. A lamp is typically seasoned prior to measuring its initial photometric characteristics. It is an accepted industry procedure for obtaining the initial rating of light sources.<sup>6</sup>

## 4.0 Physical and Environmental Test Conditions

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### 4.1 General

It is good laboratory practice that the storage and testing of lamps should be undertaken in a relatively clean environment. For lamps that operate at a high bulb temperature (highly loaded lamps), contaminants can be etched into glass bulb surfaces during operation. Therefore, these lamps should be cleaned before mounting on the life test racks.

### 4.2 Vibration

Lamps shall not be subjected to excessive vibration or shock during life testing.

### 4.3 Temperature

Ambient temperature should be controlled within the limits set by the ballast manufacturer. When the recommended testing temperature range is exceeded, life testing should be suspended.

The ballast should be operated at a temperature less than or equal to the maximum recommended case temperature.<sup>7</sup>

When specified by the manufacturer, critical lamp temperatures, such as the bulb wall temperature or the control point temperature, shall be maintained.

### 4.4 Airflow

Airflow does not normally affect the performance of HID lamps. However, special test conditions, such as unjacketed lamps operating in open areas, may require consideration of the effect of airflow.

#### 4.5 Lamp Operating Position and Spacing

The test requestor should specify the operating position of the lamps under test. Where a position is not specified, or where more than one position is specified, the lamp should be tested in the position that will be used in the application.

The lamps shall be spaced to allow airflow around each lamp. This is facilitated by designing open life testing racks with minimal structural components to block airflow.

Since the performance of HID lamps is significantly affected by operating position, it is advisable to design the life test lamp holders to be adjustable for any operating position from base-up to base-down, including orientation or tilt about the arc axis for positions other than vertical.

### 5.0 Electrical Test Conditions

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#### 5.1 Input Voltage

Input voltage shall conform to the rated input voltage (root-mean-square, RMS) and frequency of the auxiliary equipment. If the input voltage of the auxiliary equipment is a range, the center value shall be used and reported as a test condition.

#### 5.2 Line Voltage Waveform

The power supply shall have a voltage waveform such that the total harmonic distortion does not exceed 3% of the fundamental.

#### 5.3 Voltage Regulation

The input voltage shall be monitored and regulated to within  $\pm 2\%$  of the rated RMS value for the duration of the life test.

#### 5.4 Ballast

The type of ballast chosen may affect lamp life and electrical values during life. In general, the types of ballasts having the widest use should be chosen for the test. Ballasts used in life tests shall comply with the specifications in ANSI C82.4-2017<sup>7</sup> and ANSI C82.9-2016.<sup>8</sup> Supplemental equipment specified, such as starters and

capacitors for use in conjunction with a ballast, shall be used.

#### 5.5 Instrumentation

Accurate recording of elapsed operating time is important in lamp life testing. If used, an elapsed-time recording device shall accumulate time only when the associated lamps are operating. Video monitoring, current monitoring, or other means can be used to determine elapsed operating time if designed to provide sufficient temporal accuracy.

To facilitate the periodic checking of auxiliary equipment or lamp characteristics, appropriate test receptacles may be provided to permit instruments to be connected to each circuit. Instruments shall measure true RMS values.

#### 5.6 Wiring

All wiring of life test racks shall be in accordance with national, state or provincial, and local electrical codes. The ballast case and ground wire, if supplied, shall be grounded.

Attention shall be paid to the color coding of the ballast wiring layouts and wire color coding. In general, it is important to connect the "hot" leads to the center contact of the socket, with the ground lead connected to the socket shell.

#### 5.7 Lamp Connections

Lamp holders shall be of a porcelain-type construction, with sturdy threaded shells and center contacts. Lamp holders shall be rated for the specific test application under which they are used. The lamp holders used for high pressure sodium lamps and pulse-start metal halide lamps shall be pulse-rated if a high voltage pulse is used to start the lamp.<sup>9</sup> For non-screw-base lamps, the same principles of thermal and mechanical stability shall be applied to the lamp mounting mechanism.

Inspection of lamp holder contacts should be made whenever lamps are placed on the life test rack. Lamp holders are to be replaced if they show signs of weak center contacts, corroded shells, or worn insulation.

#### 5.8 Safety Considerations

Workers in the life test areas shall be equipped

with protective clothing and eye protection to avoid potentially hazardous radiation, and shall be instructed as to the possible harm of excessive visible and ultraviolet radiation, which includes severe skin burns and eye damage. The high radiance of mercury vapor and metal halide lamps in the blue spectral region is a potential eye hazard. Therefore, operating lamps shall not be examined visually without proper protective eyewear.<sup>10</sup> High-temperature fiberglass curtains are sometimes used on life test racks as a protective radiation shield.

While jacketed mercury vapor and metal halide lamps may individually emit only a small amount of ultraviolet energy, the collective radiation from many lamps at the close range necessary for lamp inspection and for changing lamps, may be large.

It is imperative that great care be exercised to avoid exposure to a mercury vapor or metal halide lamp with a broken or missing envelope or without an outer jacket. An exposure of a few seconds at close range from an unshielded arc tube operating in an open lighting unit or luminaire is sufficient to cause eye and skin damage severe enough to require emergency medical attention. A lamp with a missing or broken envelope shall be immediately turned off and removed from the life test station.

## 6.0 Lamp Test Procedures

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### 6.1 Lamp Selection

The lamps shall be selected to represent the purpose of the test. If the test is designed to evaluate specific characteristics, the sample selection needs to take this into consideration. The value of the test will depend upon the method of sampling, size of the sample, conditions of testing, and other factors. The effects of such parameters are discussed in *ANSI/IES LS-4-20, Lighting Science: Measurement of Light – The Science of Photometry*.<sup>11</sup>

### 6.2 Lamp Handling

Prior to operation, lamps shall be cleaned to eliminate handling marks, which would be etched into the glass surface under the temperature conditions encountered during life testing.

If the lamps are to be removed from the life test racks for evaluation, handling, transporting, and storing, maintaining the lamps in the same orientation as during operation on the life test racks can reduce lamp stabilization time for evaluation and for subsequent continuation of life testing. This practice is generally followed to prevent movement of the condensed metals from their normal location during lamp operation. HID lamps should cool to ambient temperature prior to being disturbed.

### 6.3 Lamp Marking

Individual lamps should be tracked during life testing. Lamps can be identified by markings applied directly to the lamps, or by labels that can be attached to the lamps during transport and evaluation or to the life test rack position occupied by the lamps during life test. The identification method selected should consider the effect of exposure to light, UV and IR radiation, and heat.

Possible suitable marking methods and/or materials include durable bar coding, ceramic ink marking, high temperature markers, or any other method which endures or can be periodically renewed for the length of the life testing.

### 6.4 Operating Cycle

The standard operating cycle shall be 11 hours on, 1 hour off. Life data from different operating cycles cannot be compared without additional information. There is no industry consensus method for accelerated life testing of HID lamps.

### 6.5 Recording Failures

Checking for lamp failures either by visual observation or automatic monitoring shall be done at an interval of no more than 1% of rated life. The recorded failure time shall be determined as the midpoint of the last monitoring interval. Each failure shall be investigated to make certain that it is a lamp failure and is not caused by improper functioning of the auxiliary equipment or lamp holders. If multiple-lamp ballasts are used, the failed lamp shall be replaced by another operating lamp of the same lamp type.

Lamp failures can be divided roughly into three classes: those caused by manufacturing defects, those caused

by normal failures, and lamps showing markedly low light output.

## 7.0 Test Report

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Test reports shall include the hours at which lamps failed for each lamp installed at the start of the test. The report shall also list all pertinent data concerning conditions of testing, type of equipment, and lamp types being tested. Typical items reported are:

- a. Number of lamps tested
- b. Selection criteria
- c. Description of lamps
- d. Description of auxiliary equipment
- e. Operating cycle
- f. Ambient conditions
- g. Input voltage to ballast
- h. Lamp operating position
- i. Hours at which lamp failed
- j. Class of failure of each lamp
- k. Lamp monitoring interval
- l. Median life of entire group
- m. Special test conditions. (Example: Minimum light output of an individual lamp before it is considered a lamp failure; any unusual conditions observed.)

## Annex A – Determining Lamp Lumen Maintenance

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It is often desirable to measure the lamp lumen maintenance during the life test. This function of lamp performance is determined from measurement of the photometric characteristics of the lamp during the life test, typically expressed as a percentage of rated output. Photometric measurements are made after at least two intervals, customarily 40% to 70% of rated life. These values are compared to the initial lamp output measured after the initial seasoning time of the lamp. (See **Section 3 Nomenclature and Definitions**.) More intervals can be used to further define the shape of the lamp lumen maintenance curve.

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